

Exam P Key Formulas

Fundamentals
$A' = A^c = A$ complement
$P[A^c] = 1 - P[A]$
$P[A \cup B] = P[A] + P[B] - P[A \cap B]$
Mutually exclusive if $P[A \cap B] = 0$

Conditional Probability
$P[A \mid B] = \frac{P[A \cap B]}{P[B]}$
$P[A \cap B] = P[A] \cdot P[B \mid A] = P[B] \cdot P[A \mid B]$
A and B are independent means: $P[A \cap B] = P[A] \cdot P[B]$ (definition) $P[A \mid B] = P[A]$ (implied)
A , B , and C are independent if every possible combination factors e.g., $P[ABC'] = P[A]P[B]P[C']$ $P[A^cB] = P[A^c]P[B]$
$A_1, A_2, \dots, A_k$ are a partition if $A_i \cap A_j = \emptyset, i \neq j$ $\sum P[A_i] = 1$
If $A_1, A_2, \dots, A_k$ are a partition: Law of total probability: $P[B] = \sum P[B \mid A_i] \cdot P[A_i]$ Bayes' Theorem: $P[A_i \mid B] = \frac{P[A_i \cap B]}{P[B]} = \frac{P[A_i \cap B]}{\sum P[B \mid A_j] \cdot P[A_j]}$

CDFs and Densities
$F(x) = P[X \leq x] = \text{CDF}$
Discrete X : $F(x) = \sum_{n \leq x} P[X = n]$
Continuous X : $F(x) = \int_{-\infty}^x f(t) dt$
Density $f(x) = \frac{d}{dx} F(x)$
Mixed distributions: If $F(x)$ has a jump at a , e.g., $F(a) > \lim_{x \uparrow a} F(x) = F(a^-)$ Then $P[X = a] = F(a) - F(a^-)$

Moments
Discrete: $E[X] = \mu = \sum_x x \cdot P[X = x]$ $E[X^2] = \sum_x x^2 \cdot P[X = x]$ $E[g(X)] = \sum_x g(x) \cdot P[X = x]$
Continuous: $E[X] = \mu = \int x \cdot f(x) dx$ $E[X^2] = \int x \cdot f(x) dx$ $E[g(X)] = \int g(x) \cdot f(x) dx$
Mixed: Find discrete / continuous pieces, then sum
Survival approach: If $X \geq 0$ then: $E[X] = \int_0^\infty S(x) dx$ $E[g(X)] = \int_0^\infty g'(x)S(x) dx$ assuming $g(0) = 0$

Variances
$\text{Var}[X] = \sigma^2 = E[X^2] - (E[X])^2$
$E[X^2] = \text{Var}[X] + (E[X])^2$
$\text{Var}[cX] = c^2 \text{Var}[X]$
$\text{SD}[X] = \sqrt{\text{Var}[X]}$
Coefficient of Variation = $\frac{\text{SD}[X]}{E[X]}$

Joint Distributions
$P[X = x] = \sum_y P[X = x, Y = y]$ = Marginal distribution of X
$P[X = x \mid Y = y] = \frac{P[X = x, Y = y]}{P[Y = y]}$ = Conditional distribution of X
$F_{X,Y}(x, y) = P[X \leq x, Y \leq y]$
$F_X(x) = P[X \leq x] = F_{X,Y}(x, \infty)$

Covariances
$\text{Cov}[X, Y] = E[XY] - E[X] \cdot E[Y]$
$\text{Cov}[X, X] = \text{Var}[X]$
$\text{Cov}[aX + bY, cZ + dW]$ = $ac\text{Cov}[X, Z] + ad\text{Cov}[X, W]$ + $bc\text{Cov}[Y, Z] + bd\text{Cov}[Y, W]$
$\text{Var}[aX + bY] = a^2\text{Var}[X] + 2ab\text{Cov}[X, Y] + b^2\text{Var}[Y]$
$\text{Var}[\sum X_i] = \sum \text{Var}[X_i]$ if independent

Conditional Moments
$E[X \mid Y = y] = \sum_x x \cdot P[X = x \mid Y = y]$
$E[X] = E[E[X \mid Y]]$

Order Statistics
$X_1, X_2, \dots, X_n$ iid random variables, $Y_1 \leq Y_2 \leq \dots \leq Y_n$ sorted version
$P[\min\{X_1, \dots, X_n\} > x] = P[Y_1 > x] = (P[X > x])^n$
$P[\max\{X_1, \dots, X_n\} \leq x] = P[Y_n \leq x] = (P[X \leq x])^n$
$f_{y_i}(y) = \text{density of } Y_i = i \cdot \binom{n}{i} [F_X(y)]^{i-1} f_X(y) [1 - F_X(y)]^{n-i}$